

Branch and Bound

Algorithm for ILP (NP-Hard)

$$\max 10x_1 + 15x_2$$

$$\text{s.t. } 8x_1 + 4x_2 \leq 40$$

$$15x_1 + 30x_2 \leq 200$$

$$x_1, x_2 \in \mathbb{Z} \geq 0$$

1. Solve Linear Relaxation: Upper bound on the optimal integer solution

In this case: $x_1 = 2.22, x_2 = 5.56, obj = 105.6$

2. Find lower bound: $x_1 = 2, x_2 = 5$ must be a lower bound ($obj = 95$)
3. partition choosing an arbitrary “branching” variable then create range $x_2 \leq 5, x_2 \geq 5$
4. Have same ILP but add the branching constraint and solve the resulting relaxations for both branches
5. Choose a node (book suggests choosing better objective value node) and continue with branching on a different variable
6. Continue until nodes become integers and until all nodes are lower than the lowerbound, highest found integer solution (lower nodes always have lower objective)

theres a bunch of heuristics with choosing which variable to bound and which node to branch